

Saif Mohammed

Systems and Biomedical Engineering Student

Egypt - Cairo | +201114313677 | ssaifmohammed04@gmail.com | linkedin.com/in/saif-mohammed-ali |
github.com/SaifMohammed22 | saifmohammed22.github.io

Education

Cairo University – Faculty of Engineering

Expected Graduation: July 2027

B.Sc. in Systems and Biomedical Engineering

Relevant Coursework: Deep Learning, Linear Algebra, Multivariate Calculus, Probability and Statistics, Object-Oriented Programming, Data Structures and Algorithms

Experience

- **Undergraduate Research Assistant, Cairo University** *December 2025 – April 2026*
 - Developing a longitudinal Multiple Sclerosis lesion tracking and labeling system for multi-timepoint MRI data.
 - Implementing medical image processing and registration pipelines using NumPy, SciPy, ANTsPy and NiBabel.
- **AI Developer Intern, Siemens DISW** *July 2025 - August 2025*
 - Designed and implemented LLM-powered features for internal engineering tools using existing LLM APIs and Retrieval-Augmented Generation (RAG) systems.
 - Built structured prompt pipelines to improve response relevance and reliability.

Technical Skills

- **Programming:** Python, C, C++, JavaScript, Bash
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, NiBabel
- **Backend & Deployment:** Flask, FastAPI, Docker, CI/CD
- **Tools:** Git, GitHub, Linux, FSL

Projects

Seq2Seq Model From Scratch GitHub | *Deep Learning, Sequence Models, LSTM*

- Implemented an end-to-end Seq2Seq architecture based on Sutskever et al.
- Built an English-French neural machine translation system trained on 34,802 sentence pairs.
- Achieved a 38.3 BLEU score with 173 sentences per second inference speed.

Alzheimer's Disease Modeling with PINNs GitHub | *Machine Learning, PDE*

- Developed a Physics-Informed Neural Network to model tau protein diffusion-reaction dynamics.
- Improved PDE parameter estimation using symbolic regression for interpretability.

MyMicrograd GitHub | *Machine Learning, Autograd*

- Implemented a minimal automatic differentiation engine inspired by micrograd.
- Demonstrated nearly 100× speed improvement of reverse-mode for neural network training workloads.

Keras OpenVINO Contribution GitHub PR #20982 | *Open Source, Python, Keras, OpenVINO*

- Implemented `numpy.dot` support in Keras' OpenVINO backend.
- Authored and merged PR improving backend compatibility and execution support.

Honors & Awards

- **NASA Space Apps Cairo Local Winner & Global Nominee** *Oct 2024*
- **4th Place - Undergraduate Engineering Mathematics Research Forum** *Dec 2024*
Cairo University - Technical Center for Career Development

Certificates & Courses

- **AWS Educate Machine Learning Foundations** *Jun 2025*
- **Introducing Generative AI with AWS** *May 2025*
- **Advanced Learning Algorithms** *Dec 2024*
- **Supervised Machine Learning: Regression and Classification** *Aug 2024*